

Ashmit Mukherjee

New York University Abu Dhabi

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Education

New York University Abu Dhabi

Expected May 2027

B.S. in Computer Science. Studied at NYU Abu Dhabi, NYU Paris, and NYU New York.

Relevant coursework: Natural Language Processing; Principles of Data Science; Statistics for Social & Behavioral Sciences; Data Structures & Algorithms; Computer Systems Architecture; Operating Systems.

Research Interests

I build and evaluate AI systems using data, statistical inference, machine learning, experimental design, and research engineering. My work spans human-AI collaboration, parameter-efficient model adaptation, and multilingual NLP.

Publications and Working Papers

Benjamin Rosche, **Ashmit Mukherjee**, and Hanan Salam. *Working paper on human-AI collaboration in shared supply chains*, June 2026.

Ashmit Mukherjee and collaborators. “Bio-Informed LoRA for Signal Peptide Prediction.” *Manuscript in preparation*, 2026.

Research Experience

Research Assistant: Human-AI Collaboration, NYU Abu Dhabi

2026–Present

- Contributing to study development and behavioral measurement for an online research project on collaboration in shared supply chains, with Benjamin Rosche and Hanan Salam.
- Co-authored a June 2026 working paper; implementation of related research is ongoing.

Research Assistant: eBRAIN Lab, NYU Abu Dhabi

2026–Present

- Implemented and evaluated **Bio-Informed LoRA**, an ESM-2 adaptation method that combines LoRA adapters with a separate projected path for BLOSUM62, hydrophobicity, and Grantham residue priors, added to hidden states before classification.
- In the original three-seed 650M comparison, BLOSUM62-guided LoRA had the highest mean across the reported primary metrics.
- Designed multi-seed evaluation protocols and a one-factor sensitivity screen showing that several single-seed improvements did not replicate.
- Manuscript in preparation (see Writing).

Research Projects

Hinglish Named Entity Recognition Benchmark

Fall 2024

- Fine-tuned multilingual transformer models, mBERT and XLM-RoBERTa, on the COMI-LINGUA dataset for Hindi-English code-mixed NER.
- Documented zero-shot baseline experiments alongside fine-tuned models, with evaluation and error-analysis code in the public repository.
- Analyzed trade-offs between model scale, task-specific fine-tuning, generalization, and computational cost.

Data Science Salary Prediction Platform

Fall 2025

- Built an interactive Streamlit prototype for exploring a 742-record cleaned salary-posting dataset, training a salary model, and inspecting feature contributions with SHAP.
- Implemented optional PyCaret and MLflow workflows for model comparison and experiment logging.

Community-Level Crime Modeling

Spring 2025

- Built an interactive Streamlit prototype that fits a six-feature linear regression to 1,994 complete cases from a community-crime dataset.
- Added exploratory visualizations and scenario inputs; results are descriptive rather than causal.

CAMP: Campus Asset Management Platform

Fall 2025

- Contributed to a team-built platform for campus resource allocation and scheduling under shared constraints, including role-based access and conflict resolution.
- Developed RESTful APIs and contributed to integration testing and GitHub Actions CI/CD.

Industry Experience

Machine Learning Engineer Intern, Zeek

Jun–Aug 2025

- Developed and evaluated segmentation models on noisy real-world data, iterating on preprocessing and model selection to assess model reliability.
- Applied model-interpretability techniques to study robustness and bias sensitivity as part of model evaluation.
- Contributed to the team's evaluation workflow by adding standardized metrics and validation procedures to the model development cycle.

Technical Skills

Languages: Python, C++, C#, JavaScript

ML & NLP: PyTorch, Transformers (Hugging Face), Scikit-Learn, XGBoost, PyCaret, SHAP

Computational Modeling: NumPy, NetworkX, game-theoretic simulation

Data Analysis: Pandas, SciPy, Matplotlib, Seaborn, Plotly

Experimental Methods: Experimental design, regression, cross-validation, multi-seed evaluation, sensitivity analysis

Research Engineering: MLflow, Git, Docker, GitHub Actions, REST APIs, automated testing, MongoDB

Applications: Streamlit, React, Node.js, L^AT_EX